

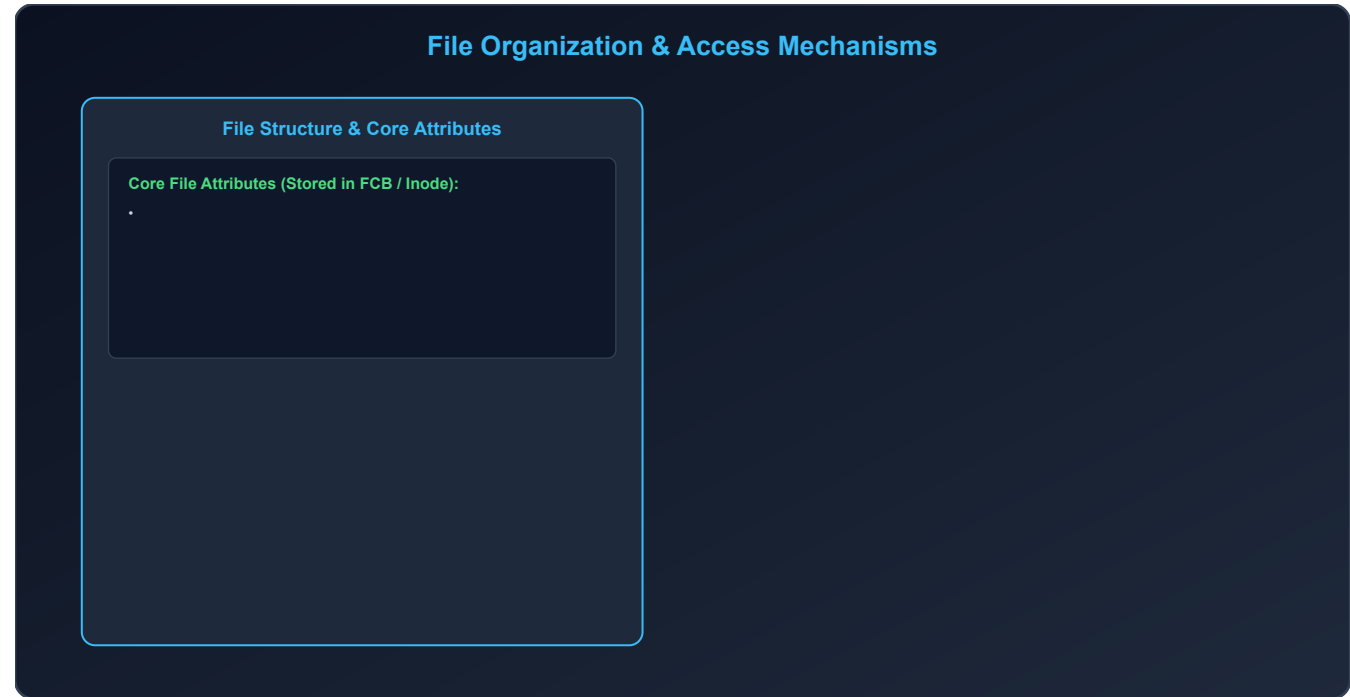
Module 07: File Concept and Access Methods

1. Core File Attributes & Operations

Files OS ke logical storage units hain jinke metadata (Name, Identifier, Location, Size, Permissions) ko **File Control Block (FCB / Inode)** me store kiya jata hai.

2. Access Methods Comparison

Method	Mechanism	Random Access Speed	Ideal Application
Sequential	One byte/record after another	Slow ($O(n)$)	Compilers, media players
Direct (Random)	Directly read block n via offset	Instant ($O(1)$)	Relational Databases
Indexed (ISAM)	Index lookup → Direct fetch	Fast ($O(\log n)$)	Enterprise account lookup



Name: Human-readable string (e.g., report.pdf) • **Identifier:** Unique integer tag (Inode number in Linux) • **Type:** Extension or magic bytes (.c, .txt, .exe) • **Location & Size:** Pointer to device & current byte size • **Protection & Timestamps:** Access rights, creation/mod time

Standard File Operations:

- `create()`: Allocate space & directory entry
- `write()` / `read()`: Transfer bytes at current file pointer
- `lseek()`: Reposition file pointer without I/O
- `truncate()`: Erase data keeping attributes intact
- `unlink()`: Release directory reference & free blocks

3 File Access Methods

1. Sequential Access (Tape / Stream Model) Information is processed in order, one record after another. Operations: `read_next()`, `write_next()`, `reset()`. Use case: Audio/video media streams, compilers scanning code.
2. Direct / Random Access (Disk Block Model) File is viewed as a numbered sequence of fixed-size blocks. Operations: `read(block_n)`, `write(block_n)`, `seek(position)`. Use case: Relational Databases, B-tree indexes, virtual disk images.
3. Indexed Sequential Access Method (ISAM) Built on top of direct access: Ek alag index table maintain hoti hai. Search index → Get block pointer →

Direct seek to record. Multi-level indexing used when index file itself becomes too large. Use case: Banking databases, enterprise flight reservation systems.